Practical - 8 Discover host using Nmap

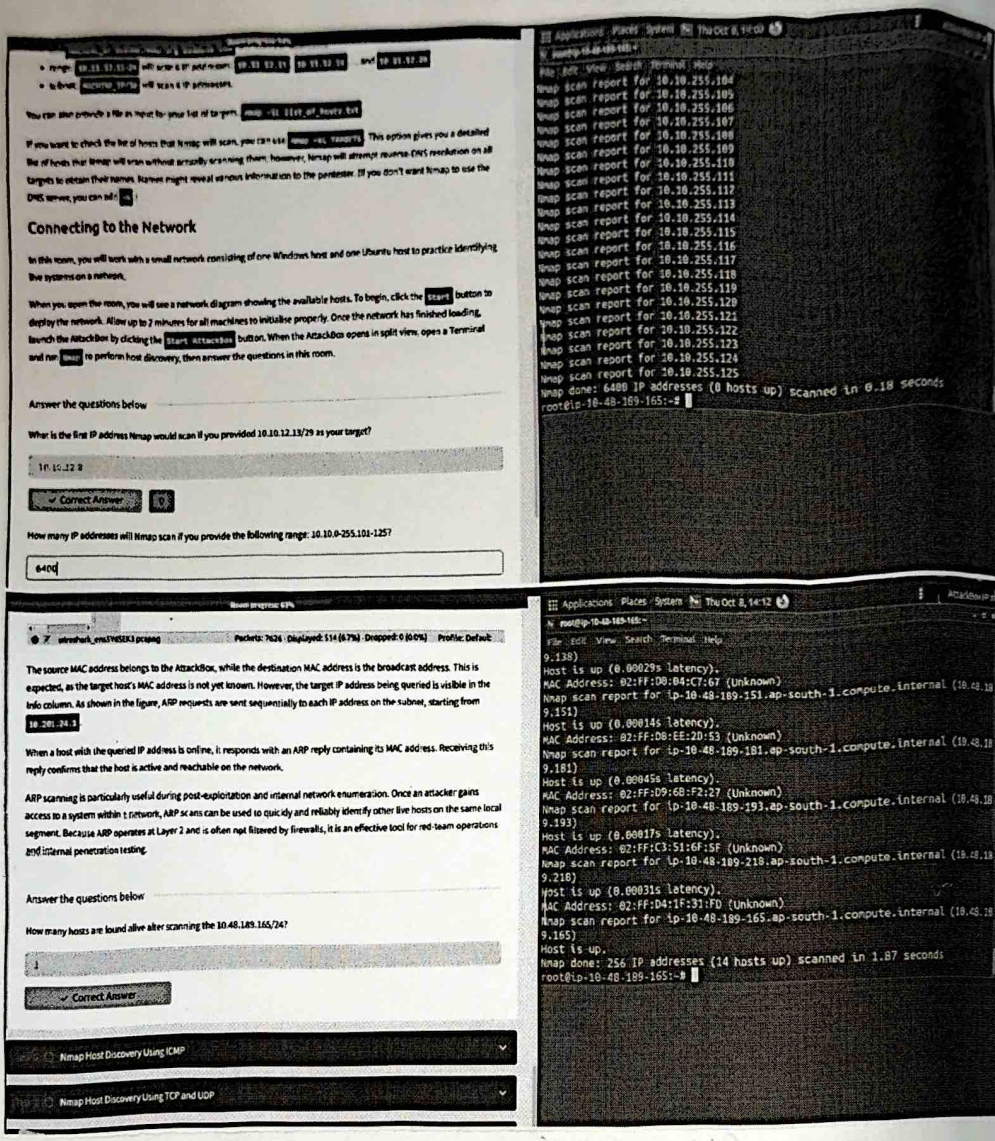
Aim; To discover live hosts on a network using Nmap host-discovery scans- ARP scan, ICMP scan, and TCP/UDP ping scan- and to understand which protocol operates all which layer of the TCP/IP model, and when each one applies.

I Introduction

Nmap ("Network mapper") is a free, open source network-scanning tool created by Gordon Lyon and first released in 1997. It is the industry-standard tool for mapping network discovering live hosts, and identifying the services running on them.

Nmap can determine whether a host is alive using four different protocols, one at each relevant layer of the TCP/IP stack

- ARP (Link layer) – for hosts on the same local subnet
- ICMP (Network Layer)- echo, timestamp, and address-mask requests
- TCP (Transport layer) - SYN and ACK Ping Probes.
- UDP (Transport layer) - relies on an ICMP "Port unreachable" reply



Connecting to the Network

In this room, you will work with a small network consisting of one Windows host and one Ubuntu host to practice identifying live systems on a network.

When you open the room, you will see a network diagram showing the available hosts. To begin, click the Start button to deploy the network. Allow up to 2 minutes for all machines to initialise properly. Once the network has finished loading, launch the AttackBox by clicking the Start AttackBox button. When the AttackBox opens in split view, open a Terminal and run nmap to perform host discovery, then answer the questions in this room.

Answer the questions below

What is the first IP address Nmap would scan if you provided 10.10.12.13/29 as your target?

10.10.12.8

✓ Correct Answer

How many IP addresses will Nmap scan if you provide the following range: 10.10.0-255.101-125?

6400

The source MAC address belongs to the AttackBox, while the destination MAC address is the broadcast address. This is expected, as the target host's MAC address is not yet known. However, the target IP address being queried is visible in the Info column. As shown in the figure, ARP requests are sent sequentially to each IP address on the subnet, starting from 10.201.24.1.

When a host with the queried IP address is online, it responds with an ARP reply containing its MAC address. Receiving this reply confirms that the host is active and reachable on the network.

ARP scanning is particularly useful during post-exploitation and internal network enumeration. Once an attacker gains access to a system within a network, ARP scans can be used to quickly and reliably identify other live hosts on the same local segment. Because ARP operates at Layer 2 and is often not filtered by firewalls, it is an effective tool for red-team operations and internal penetration testing.

Answer the questions below

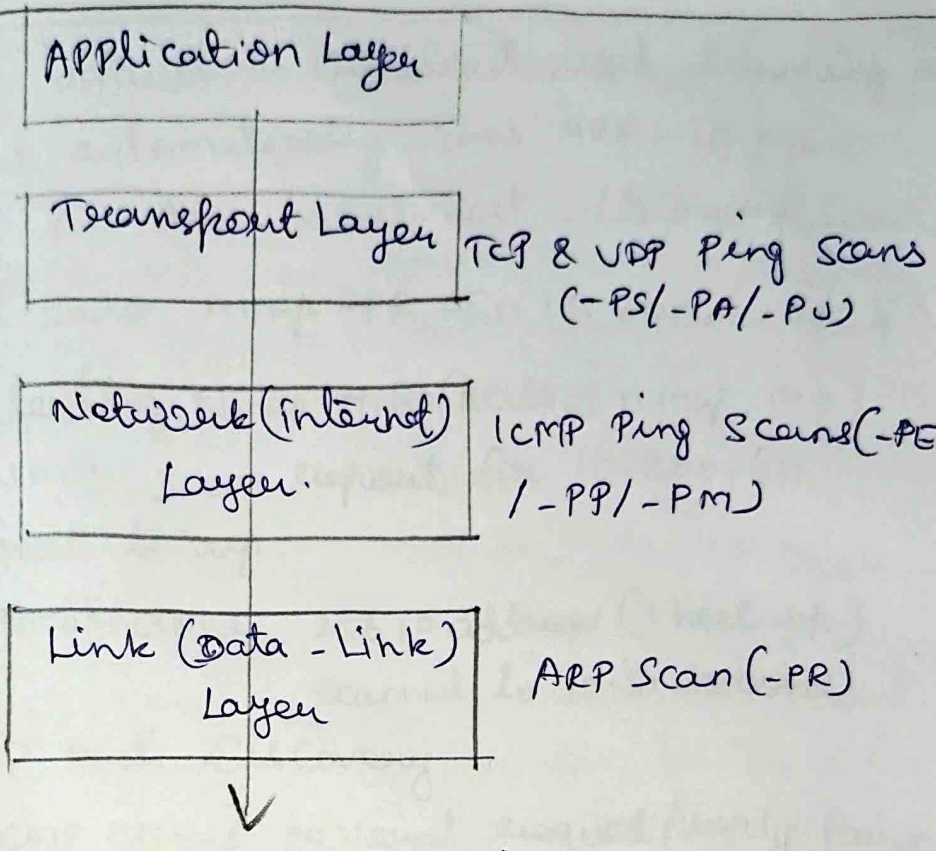
How many hosts are found alive after scanning the 10.48.189.165/24?

1

✓ Correct Answer

Nmap Host Discovery Using ICMP

Nmap Host Discovery Using TCP and UDP



Target Specification in Nmap

- A list : nmap 10.10.1.5 scanme.nmap.org example.com
- A range : nmap 10.11.12.15-20 (scans 15 through 20)
- A Subnet/CIDR : nmap 10.10.1.0/24
- A file of targets : nmap -iL list_of_hosts.txt

ARP Host Discovery

On a local Ethernet or wifi network. Communication requires the destinations MAC address. To learn it, a host broadcasts an ARP request, any host holding that IP replies with its MAC address. A reply proves the host is up. ARP only works within your own subnet since ARP frames are not routed.

By default, a Privileged user scanning a local automatically uses ARP. To run only an ARP-based host discovery scan

```
$ sudo nmap -PR -sn 10.200.6.0/24
Starting Nmap 7.80 (https://nmap.org)
Nmap scan report for 10.200.6.1
Host is up.
Nmap done: 256 IP address (1 host up)
           scanned in 10.51 seconds.
```

ICMP Host Discovery

ICMP offers several request/reply pairs that can reveal a live host. Windows systems and many firewalls block ICMP Echo by default. So Nmap also supports Timestamp and Address-mask requests as fallback techniques. Remember that on your own subnet, an ARP exchange still happens before any ICMP packet is sent.

ICMP Echo (-PE)

```
$ sudo nmap -PE -sn 10.200.6.0/24
Nmap scan report for 10.200.6.1
Host is up (0.069s latency).
Nmap scan report for 10.200.6.50
Host is up (0.070s latency).
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP address (3 hosts up) scanned in
```

ICMP Timestamp (-PP)

$nmap -PP -Sn 10.200.6.0/24

Nmap scan report for 10.200.6.50

Host is up (0.069s latency)

Nmap scan report for 10.200.6.250

Host is up (0.069s Latency).

Nmap done: 256 IP address (2 hosts up) Scanned in 8.44 seconds.

ICMP Address Mask (-PM)

This technique is the most Commonly filtered - It is normal to see 0 hosts respond, as shown below, even though earlier scans found hosts up. Always try more than one technique.

$ nmap -PM -Sn 10.200.6.0/24

Nmap done: 256 IP address (0 hosts up) Scanned in 52.14 seconds.

TCP and UDP Ping Discovery

When ICMP is filtered, Nmap can send crafted TCP or UDP Packets and infer that a host is alive purely from the fact that it responds at all - regardless of whether the Probed Port is open or closed.

TCP SYN Ping

Sends a TCP Packet with the SYN flag to a target Port. An open Port

answers SYN/ACK; a closed port answers RST - either reply confirms the host is up. Specify ports as -PS80, -PS21-25, or -PS80,443,8080. This technique does not require root privileges

```
$ nmap -PS -Sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency).
Nmap scan report for 10.200.6.250
Host is up (0.068s latency).
Nmap done: 256 IP address (2 hosts up)
        scanned in 33.88 seconds.
```

TCP ACK Ping (-PA)

```
$ sudo nmap -PA -Sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency).
Nmap scan report for 10.200.6.250
Host is up (0.069s latency).
Nmap done: 256 IP address (2 hosts up)
        Scanned in 36.64 seconds.
```

UDP Ping (-PU)

```
$ sudo nmap -PU -Sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP address (2 hosts up)
```

Nmap's Default Host-Discovery Behaviour

- Privileged user, local Ethernet target → ARP requests.
- Privileged user, target outside the local network → ICMP Echo, TCP ACK to Port 80, TCP SYN to port 443, and ICMP Time Stamp, combined.
- Unprivileged user, target outside the local network → TCP three-way handshake (SYN) to Ports 80 and 443.

Masscan - A Faster Alternative

```
$ masscan 10.200.6.0/24 -P443
$ masscan 10.200.6.0/24 -P80,443
$ masscan 10.200.6.0/24 -P22-25
```

Reverse-DNS Lookup

Reverse DNS resolves an IP address back to a hostname. Nmap Performs rDNS lookups on live hosts by defaults; R forces a lookup on every target while n disables lookups altogether to speed up the scan. Hostnames such as dc01.domain.local or mail.company.local can reveal a system's role. Use --dns-servers <server> to query a specific DNS server.

Command Reference Summary

Option	what it does	Layer / Protocol
-sn	Host-discovery only; Skip port scanning.	All (modifier flag)
-PR	ARP scan (default & faster on the local subnet	Link Layer (ARP)
-PE	ICMP Echo request/reply	Network layer (ICMP TYPE 8/0)
-PP	ICMP Timestamp request/reply	Network layer (ICMP TYPE 13/14)
-PM	ICMP Address-Mask request/reply	Network layer (ICMP TYPE 17/18)
-PS <Ports>	TCP SYN Ping (default Port 80)	Transport layer (TCP)
-PU <Ports>	UDP Ping; relies on ICMP Port-unreachable	Transport layer (UDP)
-PA <Ports>	TCP ACK Ping - needs roots/sudo	Transport layer (TCP)
-n	Disable reverse-DNS resolution	-
-R	Force reverse-DNS lookup on every host	-
-iL <file>	Read target list from a file	-

-sL	List target only, do not scan/Ping them	-
-Pn	Skip host discovery - treat all hosts as up	-

Step-by-Step Lab checklist

1. Log into TryHackme and open the "Nmap Live Host Discovery" room.
2. Work through Tasks 1-3 (Introduction, Sub networks, TCP/IP layer concepts) and answer the Simulator-based questions.
3. Deploy the lab network in Task 4 and start the AttackBox; not the target Subnet / CONNECTION_IP
4. Run an ARP scan; sudo nmap -PR -sn <Subnet>/24 and record how many hosts respond
5. Run an ICMP Echo scan; sudo nmap -PE -sn <Subnet>/24 - Compare the result with the ARP scan.
6. Run ICMP Timestamp (-PP) and ICMP Address mask (-PM) Scans; note which ones are filtered.
7. Run a TCP SYN Ping (-PS), a TCP ACK Ping (-PA), needs sudo,) and a UDP Ping (-PU); Compare; results and required Privilege level for each.
8. Optionally capture traffic with tcpdump or wireshark on the tun0 interface while each scan runs. Re

see the raw ARP/ICMP/TCP/UDP Packets.

9. Try nmap -SL <Subnet>/24 and nmap -R <Subnet>/24 to see target listing and reverse-DNS behaviour

10. Complete the room's knowledge-Check questions in Tasks 5-9, then move on to the "Nmap Basic Port Scans" room.

Result :-

This Nmap function is used to identify the vunrabilities of a live host and analyse the vunrable ports that are open in the host